

ChatGPT

SIS vs SIR Models

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sis model vs sir model

The main difference between **SIS** and **SIR** models is what happens **after recovery** from infection.

1. SIS Model

In the SIS model:

$$S \rightarrow I \rightarrow S$$

A person:

1. Starts as **Susceptible (S)**
2. Gets **Infected (I)**
3. Recovers
4. Becomes **Susceptible again**

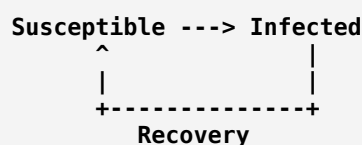
There is **no permanent immunity**.

Example Diseases

- Common cold
- Some STDs
- Certain bacterial infections

These diseases allow reinfection. [Wikipedia+2](#)

Flow Diagram



Differential Equations

$$\frac{dS}{dt} = -\frac{\beta SI}{N} + \gamma I, \quad \frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I$$

Where:

- β = infection rate
- γ = recovery rate
- $N = S + I$

Key Property

The disease can become **endemic** (stay forever in the population).

Why?

Recovered people can be infected again.

2. SIR Model

In the SIR model:

$$S \rightarrow I \rightarrow R$$

A person:

1. Starts as **Susceptible**
2. Becomes **Infected**
3. Recovers
4. Moves to **Recovered/Removed (R)**

Recovered people usually gain immunity and cannot be infected again. [Wikipedia+2](#)

Example Diseases

- Measles
- Chickenpox
- Mumps
- Rubella

These diseases often provide long-term immunity after recovery. [Wikipedia+1](#)

Flow Diagram

Susceptible ---> Infected ---> Recovered

Differential Equations

$$\frac{dS}{dt} = -\frac{\beta SI}{N}, \quad \frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I$$

Core Difference

Feature	SIS Model	SIR Model
Immunity after recovery	No	Yes
Compartments	S, I	S, I, R
Reinfection possible?	Yes	Usually No
Disease persistence	Often endemic	Often dies out
Example	Common cold	Measles

Intuition Using Real Life

SIS Example (Cold)

You catch a cold → recover → can catch it again next month.

So:

S → I → S

SIR Example (Chickenpox)

You get chickenpox once → recover → immune for life.

So:

S → I → R

In Network Science / Diffusion

In social networks and multilayer networks:

- **SIS** is used for:
 - rumors
 - malware
 - viral trends
 - diseases with reinfection

because nodes can become infected repeatedly.

- **SIR** is used for:
 - one-time influence spread
 - irreversible adoption
 - immunity-based epidemics

because once activated/recovered, nodes stop participating.

Important Graph-Theoretic Interpretation

Suppose nodes = people.

Edges = contacts.

In SIS

An infected node can:

- infect neighbors,
- recover,
- then get infected again later.

So infection may circulate indefinitely.

In SIR

Once a node recovers:

- it is removed from spreading,
- reducing future transmission paths.

Eventually, infection usually disappears.

Which One Is Used More?

Depends on the problem.

Scenario	Better Model
Pandemic with immunity	SIR
Rumor spreading	SIS
Computer virus	SIS
Vaccination studies	SIR

Scenario	Better Model
Information diffusion	Often SIR
Persistent malware	SIS

Relation to Your Multilayer Network Project

For influential-node detection:

- **SIR** is commonly used to measure:
 - spreading capability of nodes
 - final outbreak size
 - influence maximization

because recovered nodes model "already influenced" users.

- **SIS** is used when studying:
 - persistent propagation
 - steady-state infection
 - long-term survivability of diffusion

especially in temporal or multilayer networks.

Easy Memory Trick

- **SIS** → "Same again"
 - back to susceptible
- **SIR** → "Retired"
 - removed permanently



what are communities

In network science, a **community** is a group of nodes that are:

- **densely connected internally**
- but have **fewer connections with the rest of the network**

In simple words:

Nodes inside the same community interact more with each other than with outside nodes.

Intuition

Think of a social network.

People naturally form groups:

- family
- classmates
- coworkers
- gaming friends

Inside each group:

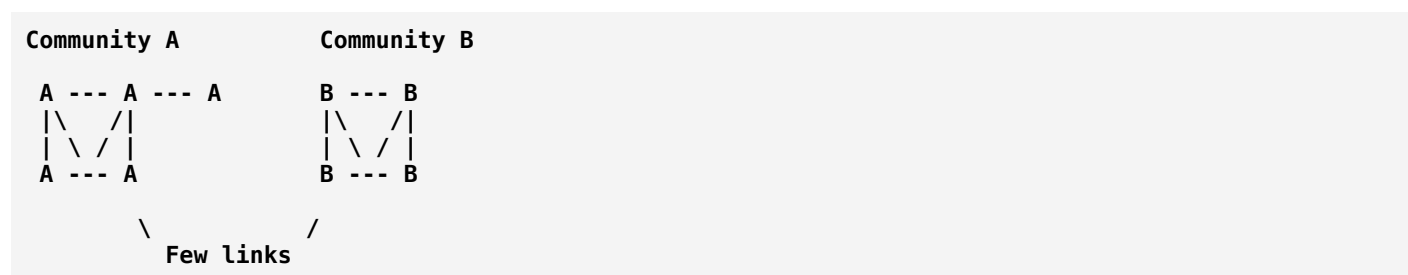
- many friendships exist

Between groups:

- fewer friendships exist

These groups are called **communities**.

Visual Intuition



- Dense links inside groups
- Sparse links between groups

Formal Definition

A community is a subset of nodes C such that:

$$\text{Internal edges} \gg \text{External edges}$$

Meaning:

- many edges among nodes in C
- few edges from C to outside nodes

Examples of Communities

1. Social Networks

On Facebook:

- school friends
- family
- office colleagues

form separate communities.

2. Citation Networks

Research papers cluster into:

- AI papers
- biology papers
- physics papers

Each field forms a community.

3. E-commerce

On Amazon:

- gaming products
- kitchen products
- fashion products

form communities based on co-purchases.

4. Biological Networks

Proteins with similar functions form communities.

Why Communities Matter

Communities reveal the **hidden structure** of networks.

They help us:

Application	Use
Social media	Friend recommendation

Application	Use
Epidemics	Understand disease spread
Marketing	Target influential groups
Cybersecurity	Detect attack clusters
Biology	Discover functional modules
Recommendation systems	Group similar users/items

Communities in Influence Spreading

This is very important for your multilayer influence project.

Suppose a node connects two communities.

That node can spread information across groups.

Such nodes are often highly influential.

Example

```
[Community A] ---- X ---- [Community B]
```

Node X :

- acts as a bridge
- spreads infection/information between communities

These are called:

- bridge nodes
- connector hubs

Very important in diffusion models like SIS/SIR.

Community Detection

The process of finding communities automatically is called:

Community Detection

The network itself does not explicitly label groups.

Algorithms discover them.

Popular Community Detection Algorithms

Algorithm	Idea
Girvan-Newman	Remove bridge edges
Louvain	Maximize modularity
Label Propagation	Spread labels through neighbors
Spectral Clustering	Use graph eigenvectors
K-clique	Find overlapping dense groups

Modularity (Very Important)

Most community algorithms optimize:

Modularity

It measures:

How strong the community structure is.

High modularity means:

- many internal edges
- few external edges

Typical formula:

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

Where:

- A_{ij} = adjacency matrix
- k_i = degree of node i
- m = total edges
- $\delta(c_i, c_j)=1$ if same community

Communities in Multilayer Networks

In multilayer networks:

communities can exist:

- within a layer
- across layers

Example

Twitter Layer

Political users cluster together.

YouTube Layer

Same political users interact again.

Together they form a **cross-layer community**.

Types of Communities

1. Disjoint Communities

A node belongs to only one group.

2. Overlapping Communities

A node can belong to multiple groups.

Example:

- student
- gamer
- employee

all at once.

Relation to Centrality

Communities affect centrality.

A node may:

- not have highest degree,
- but connect multiple communities,

making it extremely influential.

This is why community-aware centrality methods are powerful.

In Your Research Context

Community structure strongly impacts:

- epidemic spreading

- influence propagation
- diffusion speed
- multilayer connectivity

Many influential-node algorithms use:

- k-shell + communities
- entropy + communities
- bridge centrality
- modularity-based ranking

because influence spreads differently across communities.

Easy Analogy

Imagine a college.

Communities:

- CSE students
- Mechanical students
- Music club
- Sports club

A student active in multiple clubs can spread news much faster across campus.

That student behaves like a highly influential node in a network.



what is diffusion and its paremeters

What is Diffusion in Networks?

In network science, **diffusion** means:

The process by which something spreads through a network over time.

The “something” could be:

- disease
- information
- rumors
- innovations
- trends

- malware
- influence
- opinions

The spread happens through **connections (edges)** between **nodes**.

Basic Idea

Suppose:

- nodes = people
- edges = friendships

If one infected person interacts with friends, infection may spread to neighbors.

That spreading process is called:

Diffusion

Visual Example

$t = 0$
 A --- B --- C --- D
 Only A infected

$t = 1$
 A* --- B* --- C --- D

A infects B.

$t = 2$
 A* --- B* --- C* --- D

B infects C.

The infection diffuses through the network.

Diffusion Models

Different mathematical models describe how spreading occurs.

Popular ones:

Model	Meaning
SI	Susceptible → Infected

Model	Meaning
SIS	Susceptible → Infected → Susceptible
SIR	Susceptible → Infected → Recovered
SEIR	Includes Exposed state
Independent Cascade	Influence propagation
Linear Threshold	Activation by neighbors

Main Parameters of Diffusion

The behavior of diffusion depends on several important parameters.

1. Infection Rate (β)

Most important parameter.

Represents:

Probability that infection spreads through an edge.

Example

If:

$$\beta = 0.8$$

then infected node infects neighbor with 80% probability.

Interpretation

Higher β :

- faster spreading
- larger outbreaks

Lower β :

- slower spreading
- infection may die out

In Equations

$$\frac{dI}{dt} = \beta SI$$

2. Recovery Rate (γ)

Represents:

Probability that infected nodes recover.

Example

If:

$$\gamma = 0.2$$

then 20% chance of recovery per time step.

Effect

High recovery:

- infection disappears quickly

Low recovery:

- infection survives longer

3. Transmission Probability

Sometimes used interchangeably with infection rate.

Represents edge-level spreading probability.

In weighted networks:

Different edges may have different transmission strengths.

4. Basic Reproduction Number (R_0)

Very important in epidemics.

Represents:

Average number of people infected by one infected person.

Formula

$$R_0 = \frac{\beta}{\gamma}$$

Meaning

Value	Interpretation
$R_0 > 1$	Epidemic grows
$R_0 < 1$	Epidemic dies

5. Diffusion Speed

Measures:

How fast spreading reaches nodes.

Depends on:

- network structure
- centrality
- shortest paths
- infection probability

6. Time Step (t)

Diffusion evolves over time.

At each step:

1. infected nodes try to infect neighbors
2. nodes may recover

7. Initial Seed Nodes

The starting infected/influential nodes.

These are extremely important in your project.

Example

If spreading starts from:

- a random user → small spread
- a highly central user → massive spread

This is why influential-node detection matters.

8. Threshold

Used in threshold models.

A node activates only when enough neighbors are active.

Example

A person adopts a trend only if:

- at least 3 friends adopt it

9. Edge Weight

In weighted networks:

Edges have strengths.

Strong ties:

- higher diffusion probability

Weak ties:

- lower probability

10. Network Topology Parameters

Diffusion strongly depends on graph structure.

Important properties:

Property	Effect
Degree	More neighbors → faster spread
Clustering	Local trapping
Communities	Slow cross-group diffusion
Shortest paths	Faster reachability
K-shell	Core nodes spread better

Diffusion in Multilayer Networks

In multilayer networks diffusion becomes more complex.

Because spreading can happen:

- within a layer
 - across layers
-

Example

Layer 1:

- WhatsApp

Layer 2:

- Instagram

A rumor can spread:

- inside WhatsApp
- then jump to Instagram

through the same user across layers.

Additional Multilayer Parameters

1. Interlayer Coupling

Strength of connection between layers.

Strong coupling:

- faster cross-layer diffusion
-

2. Layer Importance

Some layers spread faster.

Example:

- Twitter spreads news faster than email
-

3. Layer Switching Probability

Probability that diffusion moves from one layer to another.

Diffusion Process Example (SIR)

Suppose:

$$\beta = 0.7, \quad \gamma = 0.2$$

Step 1

Node A infected.

Step 2

A infects B and C.

Step 3

A recovers.

B infects D.

Step 4

More nodes recover.

Eventually spread stops.

Why Diffusion Matters in Your Project

Your research problem is about:

Identifying influential nodes

A node is influential if it causes:

- larger diffusion
- faster diffusion
- wider diffusion

under models like SIS/SIR.

Typical Evaluation Pipeline

Researchers usually:

1. Rank nodes using centrality
2. Select top-k nodes
3. Run SIR/SIS simulations
4. Measure:
 - infected count
 - outbreak size
 - diffusion speed

Better ranking → larger spread.

Real-World Applications

Area	Diffusion Meaning
Epidemics	Disease spreading
Marketing	Product adoption
Social media	Viral content
Cybersecurity	Malware spread
Finance	Market panic
Politics	Opinion spread

Important Insight

Diffusion is controlled by BOTH:

1. network structure
2. diffusion parameters

Even highly influential nodes may fail if:

β is too low

And weak nodes may create epidemics if:

β is very high

Easy Analogy

Imagine pouring water into connected pipes.

- Nodes = junctions
- Edges = pipes
- Diffusion = water flow
- Infection rate = pipe width
- Recovery = leakage/blockage

The network structure determines how far and how fast water spreads.



what is herd behaviuor, epidemic, echo chambers

1. Herd Behavior

Herd behavior means:

People copy the actions of others instead of making fully independent decisions.

Individuals follow the crowd because they believe:

- the crowd knows better,
- or they fear missing out,
- or social pressure influences them.

Examples of Herd Behavior

Social Media

If many people share a meme, others also share it.

Stock Market

People buy a stock because everyone else is buying it.

This can create bubbles.

Real Life

- Panic buying during crises

- Fashion trends
 - Viral challenges
-

Network Perspective

In networks:

- one node influences neighbors,
- neighbors influence others,
- behavior cascades through the graph.

This creates collective behavior.

Example

```
A adopts trend
  ↓
B and C adopt
  ↓
D,E,F adopt
```

A local decision spreads globally.

Relation to Diffusion

Herd behavior is a form of:

Behavioral diffusion

instead of disease diffusion.

Important Mechanisms

1. Social Proof

People think:

“If everyone is doing it, it must be right.”

2. Peer Pressure

Neighbors influence decisions.

3. Information Cascade

People ignore personal judgment and follow previous adopters.

Threshold Model of Herd Behavior

A person adopts behavior when enough neighbors adopt it.

Example:

- adopt only if 50% friends adopt.

This is modeled using:

Linear Threshold Model

Real-World Applications

Domain	Herd Behavior
Finance	Stock bubbles/crashes
Social media	Viral trends
Politics	Mass opinion shifts
E-commerce	Product popularity
Crypto	Panic buying/selling

In Your Research Area

Herd behavior is important in:

- influence maximization
- opinion dynamics
- information diffusion
- multilayer social networks

because collective adoption depends on network structure.

2. Epidemic

An epidemic is:

Rapid spreading of disease or contagion through a population.

In network science, epidemic spreading is generalized to any contagious process.

Types of Epidemics in Networks

Not only biological diseases.

Can include:

- rumors
- fake news
- malware
- emotions
- innovations

Epidemic Process

Susceptible → Infected → Spread

An infected node transmits contagion to connected neighbors.

Epidemic Models

Model	Meaning
SI	Permanent infection
SIS	Reinfection possible
SIR	Recovery with immunity
SEIR	Includes exposed stage

Epidemic Threshold

Very important concept.

Represents:

Minimum spreading strength needed for sustained outbreak.

If spreading rate is below threshold:

- epidemic dies out

Above threshold:

- epidemic spreads widely

Epidemic in Graphs

Spread depends on:

- node degree
- centrality
- communities
- shortest paths
- multilayer connectivity

Example

A highly connected airport spreading disease globally.

The airport acts as a:

- hub node
- influential spreader

Epidemic vs Diffusion

Aspect	Diffusion	Epidemic
General spreading process	Yes	Specific contagious spread
Includes information spread	Yes	Sometimes
Biological focus	Not necessary	Usually

So:

Epidemic spreading is a special case of diffusion.

3. Echo Chambers

An echo chamber is:

A community where people are exposed mostly to opinions similar to their own.

Different viewpoints are filtered out.

The same ideas repeatedly “echo” inside the group.

Social Media Example

On X or YouTube:

Algorithms recommend similar content repeatedly.

Users interact mainly with like-minded people.

Result

People become:

- more polarized
- less exposed to opposing ideas

Network Structure of Echo Chambers

Echo chambers form due to:

1. Dense internal connections

People strongly connected to similar users.

2. Sparse external connections

Few links to different communities.

Visualization

```
[Political Group A]
A--A--A--A

      weak links

B--B--B--B
[Political Group B]
```

Information circulates mostly inside groups.

Why Echo Chambers Form

Homophily

People connect with similar people.

“Birds of a feather flock together.”

Recommendation Algorithms

Platforms reinforce similar content.

Selective Exposure

People prefer confirming beliefs.

Consequences

Effect	Description
Polarization	Extreme opinions
Misinformation	Fake news reinforcement
Reduced diversity	Less exposure to alternatives
Social fragmentation	Community separation

Echo Chambers and Diffusion

Echo chambers strongly affect spreading.

Inside Community

Diffusion is very fast.

Across Communities

Diffusion becomes slow.

Because bridge connections are weak.

Important for Multilayer Networks

Echo chambers can exist across layers:

- Twitter political groups
- YouTube recommendation communities
- Reddit ideological subgraphs

Studying cross-layer diffusion is a major research area.

Relation Between the Three Concepts

They are deeply connected.

Chain of Events

Step 1: Echo Chamber

People interact mainly with similar users.

↓

Step 2: Herd Behavior

Users imitate neighboring behavior.

↓

Step 3: Epidemic Diffusion

Ideas/rumors spread rapidly through network.

Example: Viral Fake News

1. Echo chamber forms political groups.
2. Users trust their community.
3. Herd behavior causes mass sharing.
4. Fake news spreads epidemically.

Network Science Interpretation

Concept	Network Meaning
Herd behavior	Local imitation dynamics
Epidemic	Contagion spreading
Echo chamber	Community-based isolation

In Your Research Context

These concepts are central to:

- influence propagation
- misinformation spreading
- multilayer diffusion
- community-aware centrality
- epidemic simulations

Many influential-node algorithms are evaluated by how effectively they trigger:

- epidemics,
- cascades,
- herd adoption,
inside clustered communities.